Department of Data Science, University of the Punjab,
Quid-e-Azam Campus, Lahore, Pakistan
Course Outline

Program	*BS Data Science*	
Course Code	*GE-103*	
Course Title	*Introduction to Information and Communication Technologies*	
Credit Hours	**Theory**	**Lab**
	2	*1*
Lecture Duration	*60 minutes (1.0 Hours) 2 lectures per week, and 3 hours lab session*	
Semester	*1*	
Pre-requisites	**Courses**	**Knowledge**
	None	*None*
Follow Up Courses	*None*	
Objectives	This is an introductory course in Computer Science designed for beginners. Apart from leading the participants through a whirlwind history of computing, the course also develops a feel for web programming through a series of lectures that help the students develop their own web pages. The main objective of the course is to build an appreciation for the fundamental concepts in computing and to become familiar with popular PC productivity software.	
Learning Outcomes	• Understand the basics of computing technology • Do number systems conversions and arithmetic • Have knowledge of types of software • Have knowledge of computing-related technologies	

Department of Data Science, University of the Punjab,
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Teaching-learning Strategies	• Interactive class session • Hands on practices in class • Brainstorming and group discussion sessions
Assignments	• Paper based written assignments 4 • Coding HTML and CSS 3

	Sr. #	Elements	Weightage	Details
Assessment and Examinations	1	Formative Assessment	25%	It is a continuous assessment. It includes: classroom participation, attendance, assignments and presentations, homework, attitude and behavior, hands-on-activities, short tests, quizzes etc.
	2	Midterm Assessment	35%	It takes place at the mid-point of the semester.
	3	Final Assessment	40%	It takes place at the end of the semester. It is mostly in the form of a test, but owing to the nature of the course the teacher may assess their students based on term paper, research proposal development, field work and report writing etc.

Department of Data Science, University of the Punjab,
Quid-e-Azam Campus, Lahore, Pakistan
Course Outline

Textbooks	A. Deborah Morley and Charles S. Parker, Understanding Computers: Today and Tomorrow, 16th edition, Cengage Learning, 2016, ISBN-13: 978-1337251853 B. Nell Dale and John Lewis, Computer Science Illuminated, 6th Edition.
Reference Material/Suggested Readings	C. Sinha, Pradeep K., and Priti Sinha. Computer fundamentals. BPB publications, 2010. D. Livesley, Robert Kenneth. An introduction to automatic digital computers. Cambridge University Press, 2017. E. Goel, Anita. Computer fundamentals. Pearson Education India, 2010.

Department of Data Science, University of the Punjab,
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Course Outline

Detailed Lecture wise plan

Week No.	Lecture No.	Topic	Source Book-Chapter No. (Sections / Pages)	Recommendations for Learning Activities (Mention Assignments, Test, Quizzes, Practical, Case Study, Projects, Lab Work or Reading Assignments)
1	1	**Program Introduction.** • Information Technology, Motivation, Significance and Future Prospects. • Introduction of Course	Course Outlines,	
	2	**Introduction to the World of Computers.** • Computers in Your Life. • What Is a Computer and What Does It Do? • Computers to Fit Every Need.	A – Ch. 1 (page 2 to 26)	Books Readings: A – Ch. 1.
2	3	**Introduction to the World of Computers.** • Computer Networks and the Internet. • Computers and Society.	A – Ch. 1 (page 27 to 41)	Books Readings: A – Ch. 1. Assignment#1: Review Activities from page 42 to 47.
	4	**The System Unit: Processing and Memory.** • Data and Program Representation. • Inside the System Unit. • How the CPU Works. • Making Computers Faster and Better Now and in the Future.	A – Ch. 2	Quiz#1: Topics from A – Ch. 1. Books Readings: A – Ch. 2.
3	5	**Number Systems** • Non-positional number system • Positional number system • Decimal, Binary, Octal, and Hexadecimal number system • Conversions	B- Ch. 2 C- Ch. 3	Books Readings: B – Ch. 2, C-Ch. 3.
	6	**Computer Codes** • Computer Data • Computer codes • Representation of data in binary • Commonly used computer codes	B- Ch 3 C-Ch. 4	Books Readings: B – Ch. 3, C-Ch. 4.
4	7	**Computer Arithmetic and Boolean Algebra** • Basic arithmetic operations using binary numbers • Boolean algebra Fundamentals	C- Ch. 5 C- Ch. 6	Quiz#2: Topics from B – Ch. 2, 3, C-Ch.3, 4 Books Readings: C – Ch. 5, C-Ch. 6.

	8	**Gates and Circuits** • Computers and Electricity • Gates • NOT, AND, OR, XOR, NAND, and NOR Gates • Combinational Circuits • Adders	B - Ch. 4	Books Readings: B – Ch. 4.
5	9	**Storage.** • Storage Systems Characteristics. • Hard Drives. • Optical Discs and Drives. • Flash Memory Storage Systems. • Other Types of Storage Systems.	A – Ch. 3	Books Readings: A – Ch. 3.
	10	**Input and Output.** • Keyboards. • Pointing and Touch Devices. • Scanners, Readers, and Digital Cameras. • Audio Input. • Audio Output. • Display Devices. • Printers.	A – Ch. 4	Quiz#3: Topics from A – Ch. 3. Books Readings: A – Ch. 4.
6	11	**System Software: Operating Systems and Utility Programs.** • System Software vs. Application Software. • The Operating System. • Operating Systems for Personal Computers.	A – Ch. 5 (page 174 to 187)	Quiz#4: Topics from A – Ch. 4. Books Readings: A – Ch. 5.
	12	**System Software: Operating Systems and Utility Programs.** • Operating Systems for Servers. • Operating Systems for Mobile Devices.	A – Ch. 5 (page 188 to 198)	Books Readings: A – Ch. 5.
7	13	**System Software: Operating Systems and Utility Programs.** • Operating Systems for Larger Computers. • Utility Programs. • The Future of Operating Systems.	A – Ch. 5 (page 198 to 207)	Books Readings: A – Ch. 5. Assignment#5: Review Activities from page 208 to 211.
	14	**Application Software.** • The Basics of Application Software. • Word Processing Concepts.	A – Ch. 6 (page 212 to 229)	Quiz#5: Topics from A – Ch. 5. Books Readings: A – Ch. 6.
8	15	**Application Software.** • Spreadsheet Concepts. • Database Concepts.	A – Ch. 6 (page 230 to 237)	Books Readings: A – Ch. 6.
	16	**Application Software.** • Presentation Graphics Concepts. • Graphics and Multimedia Concepts. • Other Types of Application Software.	A – Ch. 6 (page 237 to 247)	Books Readings: A – Ch. 6. Assignment#6: Review Activities from page 248 to 253.
MIDTERM				

Department of Data Science, University of the Punjab,
Quid-e-Azam Campus, Lahore, Pakistan
Course Outline

9	17	**Computer Networks.** • What Is a Network? • Networking Applications. • Network Characteristics.	A – Ch. 7 (page 254 to 269)	Books Readings: A – Ch. 7.
	18	**Computer Networks.** • Data Transmission Characteristics. • Networking Media.	A – Ch. 7 (page 270 to 277)	Books Readings: A – Ch. 7.
10	19	**Computer Networks.** • Communications Protocols and Networking Standards. • Networking Hardware.	A – Ch. 7 (page 278 to 293)	Books Readings: A – Ch. 7. Assignment#7: Review Activities from page 294 to 297.
	20	**The Internet and the World Wide Web.** • Evolution of the Internet. • Getting Set Up to Use the Internet. • Searching the Internet.	A – Ch. 8 (page 298 to 317)	Quiz#6: Topics from A – Ch. 7. Books Readings: A – Ch. 8.
11	21	**The Internet and the World Wide Web.** • Beyond Browsing and E-Mail. • Censorship and Privacy Issues.	A – Ch. 8 (page 317 to 341)	Books Readings: A – Ch. 8. Assignment#8: Review Activities from page 342 to 345.
	22	**Network and Internet Security.** • Why Be Concerned About Network and Internet Security? • Unauthorized Access and Unauthorized Use. • Protecting Against Unauthorized Access and Unauthorized Use.	A – Ch. 9 (page 346 to 364)	Quiz#7: Topics from A – Ch. 8. Books Readings: A – Ch. 9.
12	23	**Network and Internet Security.** • Computer Sabotage. • Protecting Against Computer Sabotage. • Online Theft, Online Fraud, and Other Dot Cons.	A – Ch. 9 (page 364 to 376)	Books Readings: A – Ch. 9.
	24	**Network and Internet Security.** • Protecting Against Online Theft, Online Fraud, and Other Dot Cons. • Personal Safety Issues. • Protecting Against Cyberbullying, Cyberstalking, and Other Personal Safety Concerns. • Network and Internet Security Legislation.	A – Ch. 9 (page 377 to 385)	Books Readings: A – Ch. 9. Assignment#9: Review Activities from page 386 to 391.
13	25	**Multimedia and the Web.** • What Is Web-Based Multimedia? • Multimedia Elements. • Multimedia Web Site Design.	A – Ch. 10 (page 392 to 413)	Quiz#8: Topics from A – Ch. 9. Books Readings: A – Ch. 10.
	26	**Multimedia and the Web.** • Multimedia Web Site Development. • The Future of Web-Based Multimedia.	A – Ch. 10 (page 414 to 425)	Books Readings: A – Ch. 10. Assignment#10: Review Activities from page 426 to 429.

Department of Data Science, University of the Punjab,
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Course Outline

14	27	**E-Commerce.** • What Is E-Commerce? • E-Commerce Business Models. • Types of E-Commerce Web Sites.	A – Ch. 11 (page 430 to 439)	Quiz#9: Topics from A – Ch. 10. Books Readings: A – Ch. 11.
	28	**E-Commerce.** • Implementing Web-Based E-Commerce. • Security Issues.	A – Ch. 11 (page 440 to 457)	Books Readings: A – Ch. 11. Assignment#11: Review Activities from page 458 to 463.
15	29	**Information Systems and System Development.** • What Is an Information System? • Types of Information Systems.	A – Ch. 12 (page 464 to 483)	Quiz#10: Topics from A – Ch. 11. Books Readings: A – Ch. 12.
	30	**Information Systems and System Development.** • Responsibility for System Development. • The System Development Life Cycle (SDLC). • Approaches to System Development.	A – Ch. 12 (page 484 to 501)	Books Readings: A – Ch. 12. Assignment#12: Review Activities from page 502 to 505.
16	31	**Computer Security and Privacy.** • Why Be Concerned About Computer Security? • Hardware Loss, Hardware Damage, and System Failure. • Software Piracy and Digital Counterfeiting. • Why Be Concerned About Information Privacy?	A – Ch. 15 (page 588 to 606)	Quiz#11: Topics from A – Ch. 12. Books Readings: A – Ch. 15.
	32	**Computer Security and Privacy.** • Databases, Electronic Profiling, Spam, and Other Marketing Activities. • Electronic Surveillance and Monitoring. • Computer Security and Privacy Legislation.	A – Ch. 15 (page 607 to 627)	Books Readings: A – Ch. 15. Assignment#13: Review Activities from page 628 to 631.
Final Term Exam				